ActInf GuestStream 105.1 ~ Paola Di Maio: "Systemic Deviation, A Key to E

GuestStream | *Paola Di Maio* | *1:39:26*

Hello, welcome everyone.

It is May 2nd, 2025.

We're in OctInf Guest Stream 105.1 with Paola DiMaio discussing systemic deviation from systemic failures to systemic aberration.

We will first be playing a pre-recorded lecture and then follow that with a discussion with Paola and Jeff Schulman here.

So looking forward to everyone's comments in the live chat and here we go with the pre-recorded video.

Daniel at the OctInference Institute.

I'm going to invite Daniel and friends to talk a little bit about my work.

And they picked a topic from one of my papers called Systemic Deviation.

And I've taken the opportunity to remind myself how does this systemic deviation thing I wrote about some time ago applies today.

So in the process of reviewing the paper, which I wrote a few years ago on the topic,

actually I have updated it and realized that it's not only relevant,

it also has this concept of systemic deviation.

It's got like profound implication in helping us to understand a number of things which are going on today.

And so the new title for the topic, Systemic Deviation, is from systemic failures to systemic aberration.

It's a huge topic.

So thank you so much for bearing with me.

And it's about 50 slides and I will try to go through them as quickly as possible.

But because I don't like to hurry too much and speaking too fast,

although I've got so much to say in such a little time,

I think I recommend that people listening to the talk will replay it at 1.5 speed

so that all this pausing that I do while I'm reading doesn't slow you down.

So systemic deviation refers to a behavioral and functional shift of a system from its intended purpose.

And it's particularly relevant in the context of intelligent and autonomous system.

So what is systemic deviation?

A theory, a lens, a paradigm, a framework.

Just could be anything that you want it to be in a way.

And let's take it as a book concept at the moment.

So roughly this is going to be the content of the talk.

Introduction background, examples, mitigation measures, systems.

And this table of contents in fact was written up before I finalized the slides.

So we may deviate a little bit.

This is just indicative.

And about me, I'm a teacher, writer, traveler,
interested in nature, complementary health, systems, metaphysics.
I'm asking myself the big questions all the time.
And eventually it wasn't easy for me to focus on an academic,
life because of my diversity of interest.
But then luckily in these recent years, it is okay to be multidisciplinary.
So my academic focus in the last, say, 10 years after my PhD
has been information technology, intelligence systems, knowledge representation.
I've got lots to say about this topic.
Interfaces, neuroscience, neuromodulation, mecha science.
So basically we are trying to follow and gather insights into every little topic that matters today.
So there was a paper called Systemic Deviation, the Evil in the Machine,
which can be found online easily.
And it was published by the International Society for Systems.
It was a presentation I did at some conference.
And then, um,
and it was a little bit like scratching the surface.
So this is a great opportunity.
Thank you, Daniel and Hector, who said, let's talk about this topic again,
because it's really good to go back to it and see what does this tell us about what we are at, really.
Then the other, if you want to know a little bit more, you go into psychological abuse.
And you say, why? Oh, I'll tell you later.
But basically there is an interesting,
an interesting direction for the Systemic Deviation into the
cognition and psychology of how this, what this does to people.
So, uh, thanks to another opportunity, I'm related a couple of years ago,
uh, into looking at the,
uh, psychiatry rather it was, but you know, like there was something that Systemic Deviation does to people.
And so if you may be interested in that, you can just click.
And, um, eventually what we're getting at is Systemic Abuse, Institutional Abuse,
is something that
a Systemic Deviation can help us figure out what's happening.
So why, why psychological abuse, uh, and systemic abuse?
It's because technical systems and sociotechnical systems
can be designed to perpetrate systemic abuse.
Now, you never thought about that.
When, when you, when you write down,
when you sit down to write a system,

or design a system, or implement a system,
you don't think, what is it going to do to people?
Um, you think, does it do what it's supposed to do in terms of functionality?
You think of the engineering side.
But then ultimately,
now that we are becoming so dependent on systems,
and we interact with systems at every level, every day,
um, they can be quite invasive systems,
as we are, as we are realizing with AI,
pervasive but also invasive.
So, uh,
Systemic Deviation, uh, can be turned into abuse.
When it's turned into abuse,
can lead to psychological impairment.
So ultimately, this is why,
uh, I am, uh, matching the systemic abuse,
the systemic deviation, which is more like a technical,
a technical thing,
in relation to how systems cannot do what they're supposed to do.
But it becomes relevant to social technical systems.
More importantly, and more worryingly,
that we cannot see, and cannot,
we, like users, or individuals, or stakeholders,
cannot see, and cannot understand.
If we cannot understand deviation,
we may not realize what is happening.
Between, um, the,
you know, we cannot,
it's like, it's important to understand that,
uh, systemic deviation, actually,
uh,
can have some consequences on cognition and psychology.
So, in the title,
as you can see,
the title, I put the word systemic aberration.
It wasn't there in the original work,
but now that I've had the chance to see how things are progressing,
I definitely say,

where systemic deviation will lead to systemic aberration.
What is,
what we mean, aberration?
So, actually, in science, it's,
it comes from,
a term coming from optics,
the field of optics,
and it's described as the failure of rays to converge
at one focus,
because of a defect in the lens of the mirror.
We could say that a systemic deviation is that defect,
you know,
if we use the optics analogy,
systemic deviation is like some kind of defects that prevents us from seeing,
uh,
getting the focus on,
on whatever.
So,
but also,
of course,
there is a systemic aberration,
aberration is a term used in biology.
And more generically speaking,
is the departure from what is normal,
usual or respected in a negative sense.
So,
um,
and we're seeing now how systemic deviation can lead to systemic aberration.
It's,
it's a,
it's not a,
it's not a mistake.
It's designed.
I can see that.
I've started seeing that.
And this is the main message
that this presentation gives.
I figured it out,

and I'm sharing it with people who are interested in looking into this possibility,
that we are all
conditioned by systemic deviation when we're another.
So,
before moving on,
focusing on systemic deviation,
with a bit more attention,
I'd like to share this,
table,
which was in fact made by John and I,
to which I'm grateful,
that is
comparing institutional abuse with systemic abuse.
So,
is systemic deviation leading to some kind of aberration and abuse?
Abuse is
also not particularly well understood.
It doesn't,
we don't understand how it happens and what it does to people,
but
in my work,
I've had the chance to distinguish between the two,
and,
um,
Gemini did a wonderful job at distinguishing the two using
some categories,
level,
source of arms,
scope,
focus,
perpetrator.
I don't know if I necessarily agree with this distinction here,
but it's very helpful to start as a starting point.
So,
institutional abuse,
the level is micro-meso or macro,
systemic abuse,
source of harm,

failures within,
in the institutional abuse,
the source of harm is
failures within an institution,
and the systemic abuses of broader societal structures.
And here I would put social technical systems as well.
The source of harm is the poor system.
Scope,
institutional abuse is localized,
systemic abuse is widespread,
focus,
on occurring inside the institution versus generated by the way the system is structured.
This is where our systemic deviation comes in,
really,
very clearly.
perpetrate individuals versus system structures.
So,
I think we're aiming,
we're looking at this,
but then,
very often,
the relationship.
So, when the institution is a system,
or is,
operates through a system,
then the two overlap,
clearly.
So,
the purpose of this presentation is to identify a map of issues critical to survival,
ultimately,
to health,
or healthy survival,
within the foundations of the paper,
which I mentioned earlier,
to see how this
notion of systemic deviation,
which identified in 2016,
has evolved,

and how does it apply today,
to identify further examples.
So,
in this paper,
I gave you examples,
but things have moved on,
and we have a lot more examples today.
And,
to increase situation awareness analysis,
and find possible solutions,
and find some synergy with others who
resonate with this topic.
So,
systemic failure,
in the title somewhere,
there is also systemic failure.
So,
systemic deviation will lead to systemic failure.
It is a failure that happened in a deterministic,
non-random,
predictable fashion.
This is taken from some
dictionary definition,
a Google definition.
Predictable fashion failure that happens
in a deterministic,
non-random,
predictable fashion from a certain cause,
which can only be eliminated by modification
of the design of the manufacturing process,
of the process.
Let's forget manufacturing here.
Operational procedures,
documentation,
or other relevant factors.
So,
it's very important to understand that
very often,

when things go wrong,
is that because they were designed like that.
Systemic deviation can be a contributing factor.
I'm not saying that every case is like that,
but,
and it results in a system not fulfilling its declared purpose,
or fulfilling opposing purpose,
which is,
you know,
the word the aberration takes place.
So,
to have a system that does not fit for purpose is kind of common,
but to have a system that does exactly the answer,
but designed to be,
is the blatant aberration.
So,
example of systemic failures.
Okay,
in the paper,
these examples are brought up.
I'm not gonna spend too much time on it,
because you can read the paper,
but basically,
hospitals that kill patients,
by accident.
Very common.
Corruption in law enforcement agencies.
Ooh,
very common.
States and governments run by insiders.
I'm sure you know all about that.
And democracies that are governed by,
run by elite.
You can't see that,
but you know it's there.
Media disseminating this information.
I'm sure you've heard that one.
Seemingly progressive movements that are actually set up to stir and drive repression.

This is very interesting when you see that,
especially in industry,
you say,
oh,
I didn't realize it for a long time.
You know,
as a young girl,
I was going into trade,
learning about trade and marketing.
And then I realized that the consumer association is actually driven by the industry,
or set up.
It's like,
these are,
these are,
this,
you know,
set up to deceive and to distort the consumer's view.
But,
uh,
we,
uh,
you need to evolve
the thinking and the awareness to be able to see that.
I didn't see that for a long time.
And when I did,
it did have a consequence on my ability to
function,
as in,
you know,
my psychology,
really,
and how do I perceive the world.
Open data initiative masterminded by secretive individuals.
I'm sure.
I mean,
I've seen that.
Especially the open science.
The ship's designed to sink.

I don't think that was the case with Titanic,
but I don't think it was designed to sink,
but it was definitely a design issue that set it.
And how come?
How come something like the Titanic
was not,
the flaws of the Titanic design was not spotted?
You know,
I'm not sure.
I don't think it was intentional in that case,
but it was a terrible accident.
And probably there are more in the history.
And transparency initiative designed and led by people and organizations
who want to make transparent models fit.
That's very similar to the open data.
So they're saying,
oh,
we want transparency.
And then they put the label transparency on the initiative,
but actually the initiative itself is completely opaque.
And this conflict between the label and what the thing that the system does is
cause of disorientation in anyone who's looking at it.
So when we talk about abuse,
we're looking at cause of harm,
psychological and emotional manipulation,
systemic unfairness.
Systemic unfairness is causing shifts in collective consciousness and possibly even genetic mutation.
This is,
would be interesting to investigate.
Many levels,
technical,
socio-technical,
organizational,
societal,
misinformation,
family,
workplace,
very complex,

multi-stakeholder scenarios.
So,
abuse
is not understood.
There is a lack of societal awareness
awareness of the abuse problem.
It's almost because,
as I say in the presentation about psychological abuse,
it's historical.
So,
schools,
families,
parents,
children,
relationship,
religion,
institutions,
they're all designed to
inflict some kind of abuse.
So,
where abuse is perceived as good for you,
like discipline.
Very often,
abuse,
discipline,
or good discipline is
confused with abuse.
Therefore,
we are unable to distinguish this is abuse.
Because we think that's the way things are.
And that's the way things work.
But,
I think we are trying to make progress towards a more compassionate
and mindful society.
Therefore,
we are becoming increasingly aware,
but at large in institutions,
it still ignores what abuse consists of.

And the consequences of harm are immense.
So,
what are we looking at again?
We're looking at shifting systems,
function,
behavioural structure,
causing systemic failure,
where the system does not fulfill its declared goal purpose,
or it becomes aberrated and it fulfills the
opposing goal.
And then eventually will result in systemic abuse and aberration.
So,
we're going to be looking at the mechanism,
how this is happening,
and a little bit at the outcomes.
So, the main outcome that we should be afraid of is the existential risk.
And then,
of course,
we're going to try to see if we can,
by raising or increasing our awareness,
we can attempt to mitigate this phenomenon.
So, we have defined systemic deviation.
And in particular,
I want to remind that although this presentation focuses on psychology,
and cognition,
ultimately,
its application is the design of autonomous systems.
As they become more complex,
this deviation is increasingly hidden.
And it's a little bit like what we discussed with explainability of AI,
black box,
how do we make the black box transparent,
more transparent,
how do we know what's going on inside the system,
how do we audit auditability, etc.
So, the systemic deviation applied or built into,
hidden into,
autonomous systems is relevant to complex systems,

socio-technical systems and intelligence systems.
So, why is this topic important?
Finally, we're getting to the central point.
I think, having illustrated what I'm talking about,
I can tell you why we're talking about it today.
And why I think,
I appreciate you listening.
Because people are not listening.
And that's, you know, a bit of a worry.
But, why is this topic important?
Because,
systemic deviation is invisible.
You cannot see it,
at all,
unless you really look hard.
It's covert.
It's inconceivable.
It's counter-intuitive.
And it's a mechanism that contributes to existential risks.
Existential risks resulting in technical and socio-technical malfunctions.
Loss of confidence in perception and judgment.
Mental and physical conditions.
So, it actually makes people unwell.
I think it can even cause cancer, in fact.
But, I don't have the data to back it.
Lack of awareness.
Loss of cognition.
Widespread insanity.
Madness.
You can see a lot of that.
Socioeconomic crisis.
Depletion of resources.
So, people, the resources are there.
But, they're used not to achieve the declared purpose.
They're actually wasted so that the purpose is not achieved.
And this is like,
aberration.
I'm screaming aberration here.

Depletion of resources.

Ongoing conflicts.

Disagreements wars.

So, this is...

Systemic deviation is contributing to existential risks in a number of ways,

which are very alarming.

So, when we talk about existential risks, we think of the cosmic events.

Like, the asteroid is going to hit us.

The black hole is going to swallow us.

This is not what I'm talking about.

Because this kind of existential risks are not something we...

Anyone can do about.

Climate catastrophe.

We could...

If we...

I don't know.

It's a bit late now.

Sorry.

But, it can be mitigated.

Or, we can attempt to avoid accelerating it.

If the risks are understood.

Talking about global warming and all of that.

Engineering pandemics.

Bioweapons.

Nuclear war.

Uncontrolled bio-tests.

So, we still could try to figure something out here.

But, I'm talking about...

Something which is a bit closer to each of us.

And it's the misuse of technology.

So, we have industry.

Standards.

Which are failing to address the risks.

That is an existential risk.

So, if the industry...

If the standard body that is supposed to identify and address the risks fails to do that,

that is a bit of the systemic deviation.

And the civilization apathy.

So, people are actually becoming...

Numb.

Numb to this nonsense.

People are learning how to accept the systemic deviation as the fact of life.

And it will contribute to cognitive fog.

I don't have the time to go into this.

We'll be touching on what we mean by cognitive fog.

But I'm sure you must have heard the topic.

The people aren't figuring it out.

And eventually, possibly dementia.

Loss of cognition.

Misinformation.

Must deceive brainwashing.

All sorts of things.

So, existential risks...

Not understanding...

For me, the main...

Closest existential risk to each of us.

Is not understanding what is going on.

As a result of confounding...

Obfuscation.

Confusion.

Designed.

Confusion by design.

And it means that people cannot react or make decisions appropriately to prevent...

Whatever escalation of risk.

So, the closest level of existential risk is information, knowledge, emotion, cognition.

So, it's how we...

What information we receive.

How we process it.

How we develop our knowledge.

And how that turns or stirs our emotions and impacts our cognition.

And then, of course, there's a number of other levels here we don't have time to look at.

But, okay, we'll be resulting in suffering and harm or loss of cognition or even loss of life.

So, that's...

Existential risk is the reason why we're talking about systemic deviation.

In the previous slide and in the paper, these are the main categories identified.

We have already seen this in the paper and in the previous slide.

And now I'm picking up a few more examples that have...

Where systemic deviation has become applicable from the time the paper was written.

A few years ago to die.

Which is about 10 years old now.

Now, nine years.

So, but this is a figure which I wasn't sure about.

But, you know, we said hospital.

You know, that you go to the hospital for it to be cured and then you die.

So, this is actually backed by figures.

Medical yearly death.

This is only the US.

I don't know.

I'm horrified to think what...

If this happens in the US, I wonder how big the problem is at global level.

But basically, medical error, people die of medical error,

direct misdiagnosis and adverse prescription reaction.

These are three categories of death, called by some kind of systemic deviation.

And in fact, they all belong to the category medical error.

Misdiagnosis, adverse prescription reaction.

They're all medical errors.

But they don't receive attention.

So, but when you...

There are measles, like few cases of measles, they become like a big news.

People have died of measles, these big news.

And this is a type of systemic deviation.

There are two systemic deviations here.

One, these figures show us that people are dying from the health system.

And it's showing us that information is representing...

Is misrepresenting the problem by putting a lot of attention on the small cases.

Not enough on the big causes.

So, in AI, we're looking at the machine learning.

Okay?

I'm not talking about knowledge-based AI.

Talking about machine learning, which is the biggest type of AI use at the moment.

Which is causing all this interest, because it can be very powerful.

But basically, the systemic deviation in machine learning happens when data is transformed.

You know these transformers.

You must have heard of the transformers algorithm.

And the way the data is transformed can lead to the data become completely different from what it was at the beginning.

In more than one way.

So, ultimately, the AI may result in an outcome.

The process carried out by machine learning can be false.

So, when we go and look at the standards to ensure that AI is operating safely, we see that the standard doesn't even contemplate truth maintenance.

Truth maintenance is a concept in knowledge representation, which is one of the fields I'm working on.

And I'm working on this.

I'm saying, how come the standard in AI safety doesn't mention the truth maintenance?

Well, at least this was now, we're in 1st of May, 2nd of May.

And this was true in February.

So, and I have raised the alarm.

I don't know if it has been taken.

There is a chance that it has been looked into.

But they're not attributing this to me.

Because this is what happened, unfortunately.

But I'm hoping somebody has picked up the message.

So, the safety standards in AI are not adequately addressing the risks.

So, there is a proliferation of AI safety standards that does not address critical concepts such as treatment.

And then, of course, deepfakes.

So, here, deepfakes is an interesting case.

Because it's a case where the software is designed to deliberately misrepresent the fact of the truth.

And it's built-in.

So, systemic mediation is actually deepfakes.

It's when people, so the software will produce some output which is fake.

Which is not real at all.

With a lot of examples and a lot of adverse consequences.

And it's a type of systemic mediation, of course.

So, in a moment, big drama here.

So, I didn't look at this Green Deal in the Obama era for a long time.

I heard that there was a problem with the biofuels.

And I said, what's the problem with biofuels?

What problem can there be with biofuels?

And the problem is that they are,

entire government programs are adopting biofuels, which are clearly not right.

not biofuels at all.

Like, biofuels, the process of generating biofuel is supposed to be

environmental friendly and creating free energy or is like non-consuming fossil fuel.
But, in some examples, and this was designed to happen like this,
they made the biofuel dependent on fossil fuel.
Because they designed it wrong.
And then they made the administration responsible for that.
When the administration was actually probably a victim of an industry lobby that nominated the administration to sign off and accept a program which was clearly not working.
Because nobody in the program had control of what was happening.
And nobody had enough competence to say this is wrong.
So, the public could only see that it was wrong at the end.
After billions were wasted, after biofuel became a joke.
And after we realized that somebody in the system is making biofuels fail,
and is making the political administration just trying to go green fail,
and make it look ridiculous.
Because they're really evil.
And they can do that.
They have the power to do this.
And then, this is another example of the climate change conference in Brazil,
where they're cutting the trees down to make room for the climate change meeting.
This is, I don't know if you've heard of this, but this is,
I just thought, oh my god, you know, this is really happening.
So, they have the climate change conference, and they're taking down the forest to accommodate.
So, this is a case of systemic division.
For me, it is.
It's dramatic.
And so, this was an interesting example.
I was reading the papers the other day, The Atlantic.
And I don't have personal quibbles with anyone.
But they were saying, you know, here, the people who are losing this ability to figure out what's happening, they're not just the common folks like you and I.
They are the people who are supposed to be in charge of big decisions.
So, this article was telling of, you know, the world's largest military was enveloped in a level of dysfunction that bordered with the comical.
Now, we have the top brass who is being misled.
So, think of what happened with Iraq.
One simple meeting by Powell triggered years of war in the Middle Eastern War.
Weapons of mass destruction that weren't there.
So, here we have, I'm not talking about, I don't know about this guy.

I have no idea who this guy is, whether he's fit or poster or nothing.

But, what I know is that people who are making the big decisions,

these are supposed to be the most rational, best informed decision makers,

who have big power to destroy the universe,

or not to destroy the universe, and they are confused.

They're dysfunctional, they're misled, they're taking for a ride.

This is existential risk.

And then we're looking at psychological abuse in general, and mind-wracking.

Now, when somebody told me this is mind-wracking,

I couldn't figure out what we're talking about.

And then, and then I figured it.

So, when we are misinformed, and when we are given two versions of the truth simultaneously,

people, cognition and consciousness, is become split.

And with a split mind,

it's not, you know, we are not able to make the best decision at all.

And this psychological abuse, this form of psychology, this is a psychological abuse.

So, it's happening in, it's happening in the

organization, family, schools, but it's happening at government level.

So,

if somebody figures it out, and says, okay, I worked it out, this is happening.

They're taking us for a ride.

This system is designed wrong, is designed to fail.

Now, when an expert will be able, will see it, they are psychologically abused.

So, the people who identify the systemic deviation become harassed.

At best, they become psychologically abused and harassed.

At best.

At worst, they get murdered.

So, the systemic failures in Boeing 747 are historical.

There's not secret.

There is, if there are a lot of reports, for decades, there's been issues with some safety.

I'm saying Boeing, because it's the largest manufacturers, but it could be,

there could be others.

So, there are systemic failures in the safety of certain systems.

For example, Boeing 747.

And after 747 came running into accidents,

caused by safety issues repeatedly,

increasingly, waringly, systemically,

the engineers, they started shouting, saying, I told you so.

I have reported this.

My report has been shoved under the carpet.

So, people may have noticed that, you know, there are safety issues,

or some systemic deviation taking place at system design level,

or system manufacturer in some cases.

And they were killed.

Like, this guy died accidentally by suicide.

It's a very long story, but you have to read it.

And it's unbelievable.

Joan Barnett, Joshua Thin, they're now suing.

But it's dramatic.

So, when we find out that there's something going on,

and we say it,

a whistleblower, or as a simple, naive observer,

point the finger and say, this is wrong.

People will become abused at best, and they will become murdered.

And, when there's enough at stake.

And this, you know, there is some lots of stuff.

There are some examples.

And then, I'm looking at forensics.

I have a paper out, which I can share with you.

Interesting stuff, but ultimately, we have...

Forensics is sometimes used to lead to wrongful conviction.

So, we are using the most rigorous process for fact-finding

to actually produce, fabricate false evidence.

Very common.

So, why is this happening?

Motivation for systemic deviation,

systemic failure can be found in...

So, some of it are genuine errors.

Okay, we accept that humans are not perfect,

that systems, organizations are prone to failure.

So, we're okay.

But, they can also be systematic.

So, someone has figured out that they can play God

by injecting a systemic deviation into a system,

an organization, a program, a hospital, a tool,

a system that can make money out of the game.

Which can mostly be money.
But, it could be a number of other things,
including political power or personal power.
Power, money, profit.
So, people are using some...
Some, this...
The more knowledgeable people who have figured out how to control a system...
These are the system control people.
They know how to control a system by designing it wrong.
And then, they gain control or wealth or other whatever from this failure.
I would say so.
How can it happen?
Okay.
So, the mechanism are misrepresentation.
It's lying and confounding.
By confounding here,
I don't mean the statistical confounding factors that you find in statistics,
which are mathematical.
Confounding with data that doesn't belong and it makes the analysis of the data skewed.
So, the bias.
I'm talking about the tactics to confound.
So, deliberately seeding, misleading, decoying signals into a message.
So, that it takes a lot more intelligence and processing ability to decode the truth.
To clean out the truth.
What's happening really.
Poor system design.
Accidental.
So, sometimes it's accidental.
Sometimes it's deliberate.
And we need to figure out.
Or both.
We need to learn how to distinguish.
When is this happening by accident?
Or what is someone actually doing it?
So, it happens to cognitive dissonance.
So, when it's injected, it's because someone is designing a response into the individuals.
Very often, these are the decision makers.
People, the influencers.

People who can call the shots.

They are deliberately misled to have dissonant cognition.

And this is where a lot of the people in power are actually psychopaths.

Which is kind of very worrying.

Leading to alienation.

Completing emotion.

And then, of course, more general level.

These conflicts in society can lead to mental health issues.

Autism.

Depression.

Disengagement.

People say, okay, sorry, I can't cope with this.

And I see you all on the beach.

Systematic aberration.

So, this is happening.

Day in, day out.

Everything that we see, or we do, or we perceive is leading or going the same way.

And then, we become confounded.

And I suspect it can cause mutations.

I have reason to believe that.

So, there is a paper.

There is at least one paper.

There are probably more.

But it's saying mutations in human accelerated regions.

In the brain region.

So, mutation

in humans disrupt cognition.

Okay.

There is an article with study.

So, how mutation can impact cognition and social behavior.

Disrupt.

But, what I'm looking at is the opposite.

How can the, can cognition and social behavior disrupt

the cause or impact mutation?

I think it can.

But I don't have the data for that.

That's another possible direction for study.

So, but what we know for a fact is that depression is leading to health decline.

Chronic stress and depressive life behaviors associated with impairments of neuroplasticity.

People lose their ability to think creatively or positively sometimes.

Optimistically.

People lose the ability to think all together.

When they are stressed or depressed.

So, all of this systemic deviation which is causing confusion.

People are losing their ability to control their own mind.

Their own cognitive processes.

They don't understand what is going on.

They are all at the receiving end of someone doing something which obviously doesn't work or is wrong.

And that's very depressing.

And this is happening quite a lot.

It's a little bit, well, people are experiencing this cognitive fog.

And it's something that we should learn how to recognize a bit more.

Still dissonant consequences.

Dissonant psychological emotional dissociation.

So, we have a presentation on free will.

We all become conditioned to function in a way that is reducing, minimizing harm and minimizing and maximizing reward.

So, ultimately, that's where the conditioning is taking place.

So, I have a presentation on free will.

Which I can share if people are interested.

And then, of course, that's where people become manipulated and unable to realize that they are being misled into something, into being part of something bigger.

Which is definitely wrong.

This is how people become like the, you know, the horde behavior.

Because not there, because of the uncertainties, there is certain safety going where everyone else is going.

And individuals, very few people have this ability to call out this phenomena of wrong.

And when they do their risk, they become ridiculed and they become abused.

And they even murder and their families.

So, it is easier to say, okay, you know, I've got children to think about.

Oh, I have heard so many times people are, we want to worry about the end of the month paying the rent.

And sorry, we cannot cope with this.

We have to just go with the flow.

So, psychological and emotional trauma.

So, this is where I think it's happening that we are mutating.

This is all happening.

And we are mutating.

Possibly cause of cancer.

Possibly we are becoming chimpanzees.

I mean, you are becoming chimpanzees.

They are becoming chimpanzees.

I'm not, I'm trying to develop my human discrimination at some cost.

But, so basically this trauma and this trauma are impacting evolution.

Okay.

That's important thing to remember as existential risk.

So, what can we do?

We can first try to figure out what's going on, develop, raise awareness, recognize the risk of deviation.

And become paranoid in the process.

And become, how do you say, fairly

And become, very, petulant like.

You know, we have to pay attention to every little detail.

Because deviation can be very subtle.

So, you need to be able to look at the details.

And try to see, is this little thing going to lead to a bigger problem later on down the line.

Knowledge representation.

I'm sure I've got more slides on that.

So, how can it be mitigated?

Logical integrity.

Logical integrity.

We need to be able to exercise our intellectual discrimination through logical integrity.

Whatever that is.

Whatever way you define.

Truth preservation.

Integrated systems design.

Codal loops.

Stepping away.

This is more like a behavioral decision-making.

When things are wrong, we cannot understand them.

Walk.

Walk.

As fast as you can.

Walk away.

And sometimes, we find ourselves into some things.

You know, into the juggle.

We say, I had to walk away from this.

I had to walk away from that.

But, eventually, we will meet others who have walked away.

And develop better.

And maybe we have a chance to reconsider and make an impact for the future.

And develop better systems, obviously.

So, this sort of causes loops.

For those of you.

It's a design way of somewhere here.

There is a cause of the cause for the system to work as intended.

Or the system to work not as intended.

Somewhere here.

And here is where the systemic deviation will hide.

As you can see.

You need to apply a lot of attention to see where the deviation is taking place.

So, in a previous...

This is coming from other presentations.

And basically, when I say knowledge presentation, it's important.

I have reason to do so.

I've done a lot of work.

But, ultimately, in order to understand what is going on.

So, how to maintain logical integrity.

We need to be able to identify concepts.

To have an idea of how everything sticks together.

What things are called.

Glossaries.

How these entities or things relate to each other in a process.

We use frame in AI.

So, we have these.

And how they build a process or a function.

Algorithm.

And how these are written up.

So, which syntax, notation, language.

And then, ultimately, how these are encoded into a system.

So, we need to learn about knowledge presentation.

Because this is how the system is going to...

By using a correct, adequate, valid, transparent knowledge presentation.

That's the only way to figure out how the system is going to play out.

Then, we have knowledge presentation as a shared mechanism to ensure
the conceptualization of complexity.
Logical and functional integrity.
Risk reduction.
So, we were looking at our risks.
And this is the only way we can ensure to reduce the risk.
Is by understanding and implementing knowledge presentation correctly.
So, to address the black box and the opacity.
And to improve reproducibility and predictability.
So, mitigation.
Develop awareness.
Apply intelligent reasoning.
Devise better systems.
Develop cautionary measures.
Sometimes non-action.
So, whenever we see that whatever we do is misused.
Or misinterpreted.
Or twisted.
Or mischaracterized.
Misrepresented.
Then, it's better just to shut the heck up.
And walk away.
Which is why many of us are ending up in some kind of monastery situation.
Or retreats.
Or isolation.
We need to collect ourselves.
And figure out if everything that we do can be used wrong.
Because there is a bigger evil out there.
Then, sometimes we take this spiritual.
You know, I'm going to go to the beach.
Or become an artist.
Or just rather not do anything.
Which is a lot of people actually are taking this path.
And I understand why.
So, but of course, if we can.
We should design properly.
We should develop ethical principles.
We should overcome our cognitive limitations.

Understand and address systemic challenges.
So, this is the system level.
We need to.
I have a lot of papers on this.
We need to understand the system level.
And we need to do some joint optimization.
And I have a paper on that.
Which is something that happens naturally.
In functioning.
How do you say?
Well balanced ecosystems.
Joint optimization is a natural function of a healthy ecosystem.
But when this joint optimization is prevented by the wrong design.
Then equilibrium.
Self-equilibrium.
Or self-correction cannot happen.
They're just discussed in a different paper.
So, measures.
Again mitigation.
Refined design principles.
Strengthening protocols.
Enhanced awareness.
Address misuse.
Risk of misuse.
So, the joint optimization there in Portland.
Because the joint optimization means that the system should be designed to figure itself out.
Self-correct.
But in the system-activated way.
Sometimes the ability to self-automize is prevented completely.
And this is done by some time.
By bureaucracy or rules.
So, very interesting stuff.
And a very big topic.
So, purpose, structure, behavior, function.
Technical system.
Sociotechnical system has the human aspects.
Like language.
Cognition.

So, you can.

We can design technical systems.

In a deterministic way.

But we cannot determine.

How humans are going to interpret.

Or react.

Or adapt.

So, you know.

We have some knowledge presentations.

Somewhere in between.

There is again.

These are slides from other talks.

And bias.

So, I think systemic deviation is currently seen as a form of bias.

Where all of these factors.

Systemic bias.

Epistemology.

Cognition.

Algorithms.

Function.

Representation.

They all come together.

To develop the functionality.

But they can also equally be used by those very smart folks.

To develop this function.

And this is what we're seeing a lot.

And this is what I'm trying to put across with the talk today mostly.

So, we're logical chains.

So, now we're looking at.

We're taking all this stuff together.

We have misalignment.

Misrepresentation.

Cognitive dissonance.

Loss of cognition.

Disengagement.

Decline.

Ability to take your activation.

Possibly leading to catastrophe.

We are trying another way of looking at this.

Navigating all this information.

Is looking at cognitive dissonance.

Psychological trauma.

Genetic imitation.

Leading to aberrations.

Possibly we're becoming monkeys.

Humans becoming monkeys.

We're seeing that quite a lot.

And dissonance.

Leading to dissociation and dysfunction.

Some of it is visible very clearly in the health.

Human social health.

Poor system design.

Can lead to logical disintegration.

Big alert here.

Alarm.

Be careful.

Let's be aware this is happening.

And this is a view of complexity.

Again part of another piece of work I did.

And we're trying to break it down.

Where is the complexity?

People are saying things are becoming more complex.

Are they?

And here I was breaking down.

What is complexity made of?

Perceptual.

Cognitive.

Representational.

Epistemic.

Ontological.

Functional.

And systemic.

So there is.

This is what makes up our complexity basically.

And I'm talking about this somewhere else.

And again I place some emphasis on knowledge presentation.

And I identify some levels of knowledge presentation.

Corresponding to as many levels of functionality.

Other levels.

Implementation.

Logical.

Epistemic.

Conceptual.

And so.

System level knowledge representation.

You can look up the works.

Roughly consisting of cognitive and functional areas.

And I think systemic deviation is here.

But it could possibly also relate to here.

So I don't know.

I'm going to maybe redo this diagram.

I don't know where I got this tree from.

But the systemic deviation could well be here.

So I don't know.

I don't know.

Research agenda.

Knowledge presentation more.

For the reasons that I discussed.

Ethical models.

Systemic.

Addressing systemic challenges.

And a lot to be discussed of where this is going.

So misalignment between intent and action.

Can be designed.

Can be accidental.

But it can be designed.

To trigger failure.

This is what we should be very aware.

And very afraid of.

Systematic.

Misrepresentation.

Intentional misaligning.

When designed and engineered properly.

Systems can be wonderful.

They.

Self-equilibrials.

Self-correction.

Evolution.

Intelligence.

We all love this.

But then.

When you leverage this ability to.

Cure.

And point it into the right direction.

Of harm.

Then.

You know.

Something to be obviously.

Recognized and avoided.

And they're masked.

When things are masked.

By misleading labels.

And post knowledge.

So we're saying.

This is so and so.

But actually.

You won't look.

And it ain't so.

At all.

And so this is where.

Unless people are paying full attention.

To the phenomenon.

They will take.

The label.

For good.

I mean.

To.

Because we don't have the ability.

To process unlimited amount of information.

We are tired.

We are busy.

We are.

Our attention is elsewhere.
In most days.
On the children.
On the survival.
On the paying the bills.
On other things.
Which are.
More rewarding.
And more.
Like even.
You know.
Everything that makes us happy.
We are looking.
Paying attention to.
And all these misleading labels.
Ah.
You know.
It's a headache.
So to what extent.
Can we actually go and check everything out.
And split every single half.
And that's what I'm afraid.
We have to.
Try to do more.
So.
Because when that.
Happens.
There is the.
The divergence between the label.
And actually the fact.
Doesn't.
When the label doesn't correspond.
To what the label describes.
At all.
That's where.
Systemic condition takes place.
And it's used in adversarial taxes.
So it is a.

It is a weapon.

Actually.

And it's a weapon.

And the first.

Level of.

Target for the weapon.

Is the people's cognition.

And consciousness.

is the devil's ail

So there could be some short-term gain in the long term.

Everybody's gonna lose out.

And then of course, this is another big topic,

deviated intelligence.

So we are using entire systems, organizations,

rule sets, connections, networks,

to produce intelligence,

which is actually designed to mislead and dysfunction.

It's wrong, deeply wrong.

The paper on forensics, which is gonna come out shortly

with Taylor Francis, it was chapter of a book.

I have the opportunity to delve into deviated intelligence

from a forensics perspective,

and I'd be happy to share the paper with anyone interested.

So we're looking at evil, okay?

If someone was wondering, or couldn't see,

I couldn't see evil for a long time.

I thought everything was good, you know?

And now I can see it.

And I see that if you're asking me what is evil made of,

it's, this is the stuff that evil is made of.

And there is probably a bit more, but this is more or less.

So there is no right or wrong, but there is systemic deviation.

We've looked at the examples, mitigation,

ethical considerations, and research directions.

And thank you.

So yeah, I haven't managed to put in all the references.

I'm struggling to meet my deadline.

The deadline was, let's try to get this done by the 2nd of May.

Today is the 1st of May.

Putting together the recording just to make sure we've got something to talk about tomorrow.

And I hope this thing has been recorded.

I want to thank Michael Walker, who introduced me to Hector.

Hector introduced me to Daniel.

And now I think there is another guy here whose name I don't remember,

who just came on board yesterday.

And I hope everyone is going to look at this talk on YouTube,

on the Active Influence Institute.

And I am also trying to do a couple more slides specifically on the Active Influence before the 2nd.

So thank you so much for listening.

Do get in touch.

Let's talk about this stuff.

Let's talk more about this and see what can we do.

All right.

Thank you for the recorded presentation, Paola.

Are you back?

Yes, I am.

Thank you so much for listening.

I'm sorry.

So what did you do wrong about the slides not being very visible?

I don't know.

They seemed fine to me.

Oh, you see, fine.

Because when I looked through the YouTube channel, the slides were out of focus.

But Jeff, could you manage to read the slides from YouTube from the live stream?

Yes, I was able to just fine.

Oh, that's great.

Oh, that's great.

For some reason, my YouTube doesn't show.

But thank you so much for listening.

And I would like to perhaps suggest that we also link the original slides or recording to

the YouTube so that anyone having difficulties like I did, they can watch the original recording,

which seemed to be clear enough.

So thank you so much for listening.

Cool.

Yes.

What do you make?

What do you make about this?

What do you make of this?

Tell me.

Jeff, want to go first or I can go?

Yeah, I can start.

My first thought is about perspective and triangulation.

Um, looking at the first bit of slides, actually, I took some notes here, let's see.

Jeff, want to go first or I can go?

Yeah, I can start.

My first thought is about perspective and triangulation.

Looking at the first bit of slides, actually I took some notes here.

You know, from the first point of view, you talked about explainability.

There are some tools that are out there from a modeling perspective that we can look at using, like SHAP and LIME for certain datasets.

I think the conversation about deepfakes definitely warrants, you know, expanding on.

One of my research areas is dark patterns, which are deceptive user interface designs that people who create interactive systems intentionally place in these systems in order to take advantage of their users, primarily for monetary profit, but for other reasons as well.

How about we start with that?

Yeah, yeah, yeah, yeah.

Do you want to talk about it?

Yeah, yeah.

So dark patterns, dark patterns are very, very widely used.

They have significant economic impact.

There are FBI, there's an FBI report that came out, billions of dollars that are lost, and time lost as well.

So the dilutive effect of people's attention being split is very corrupting in terms of people's ability to engender meaning in their lives.

And that's, you know, we're working on a user study right now to see how dark patterns affect the elderly population, you know, retirement age adults.

I hear some clicking, some keyboard clicking.

Daniel, is that you?

No, Paola, please mute.

Paola, please mute when you're typing.

Sorry.

Sorry.

Let me mute.

I've got some Morse code coming through.

Yeah.

Yeah.

So dark patterns, they divert your attention away from what you could be offering.

Optimizing.

And we see dark patterns all around us and we see deception in nature all around us as well.

And it's not unusual.

Deception is, can often be a feature in survival.

Zebras have stripes.

You know, many insects use mimicry.

And deception necessarily isn't a good or bad thing.

It has to do with the intent and the will of the participants in that behavior.

That being said, we should absolutely optimize for well-being and wholesomeness and restoration wherever we can.

Another thing that I just took as a note here was your integration of optics.

That is very interesting to me.

I've also been considering how the evolution of language and sight are essential to our embodied experience and how gravity curves space-time, for example.

And how sound travels at a different speed of light and how we evolved, you know, like as life on earth.

We evolved touch and then sight or then, excuse me, and then, and then hearing.

And then, you know, smell and taste.

And then finally sight.

So that is, I think, something where there might be a correlation that can be explored.

From the perspective of control systems and deliberate sabotage within a system, we, I just submitted a paper, actually 24 of us just submitted a paper to a venue that's exploring that.

I believe that there is a way forward.

And it's going to be very Kahnian in how we experience it.

But I'm very hopeful because it's a bright future.

Something else that I wrote down was that, you know, crucibles have dual functions and, you know, they both, they both provide the form and the heat and they burn off the, the, the dross of the, in the refinement process.

So reframing some of these things as a more of a distillation could provide another perspective that might open up the door for alternative considerations.

So, you know, with, you know, with cognitive pruning, we have, we have, we must model and embody and enact our experiences so we can make optimized choices.

So having, you know, having a system where you're allowed to make mistakes or perceived mistakes that are really just life lessons with guardrails is healthy.

Um, what do you think Paula or Daniel?

I, I just laid out a couple of things there for the last 15 minutes or so.

Thank you.

Thank you.

I, I, I appreciate and agree and would like to look more into everything that you said, but I didn't understand about the last point, the healthy guardrails.

Could you explain what you mean there a bit more?

Absolutely.

I understand you properly.

Absolutely.

So I think this goes in line with the, uh, POMDPs, the partially observable Markov decision processes that we, um, use to model agents in active inference.

And when you're in a domain, an environment and, oh, you're clicking again, we're getting some more morph code.

You're sending secret signals.

Um, excuse me, Daniel, where was I at?

POMDP in active inference.

Oh, POMDPs in active inference.

Yes.

Oh, uh, partially observable Markov decision processes.

Um, when you.

Situate yourself in a domain and.

You have a boundary that you're doing an information exchange on with.

That environment, you cannot, you are not aware of the environment.

That domain is nested in.

So when you have.

Um, uh, a second order connection.

It can cause confusion.

If you don't have a translation mechanism to be able to.

Exchange information with, with a domain that you're not primarily connected to.

Um, okay.

Okay.

But why.

Okay.

This is a, this I understand that.

Um, I would like to, um, argue.

So to say that.

Uh, uh, a user of a system who has a second order connection with it should be made aware of this hierarchy is a matter of ethics and transparency.

So that's.

That is a compression problem.

Um, mathematically if I'm nested in a domain.

That, uh, for example, I, I am a domain of, of one.

Um, I am nest myself in a grid of, uh, 10 by 10.

Right.

And then I nest myself in a grid of a million by a million.

If I, if I extend into that second domain without first salt completely solving or solid sufficiently solving that first domain, it's going to disorient me.

And I'm going to be confused.

And I wonder if that maybe isn't the cause of some mental illness is that you don't have a trip.

You don't have a trip.

Oh, go ahead.

Yeah.

It surely is.

So I can see why you're framing the, um, this challenge, so to speak, um, mathematically and, but from a systems, uh, uh, design viewpoint that, uh, it needs to be for me, the way I stand, unless it's adversarial, unless it's designed to confuse the user for whatever reason, then it should be, the setup should be declared.

It should be saying, okay, this year up, you are here and these are the levels you're operating in.

And now what you're looking at is a level one and that's where you are.

And then you see, if you don't do that, you're, I think, abusive psychologically the user.

And so this is exactly the point that I insist on.

I'm not necessarily a step you or others agree depending what kind of system you're designing.

Um, but I, you know, I'm glad you're, you're picking up on the point exactly.

And that is where I think there is a difference in the sense of what kind of system you're designing.

I'm designing an adversarial system that you want to figure out how the user is moving, how they're thinking.

You're trying to learn what process going into the user's brain by putting them into a grid that they don't know anything about

and they have to figure it out and that you can see how the user is moving and how they're thinking or perceiving.

Um, they make this kind of ail

experience, the embodied experience should be commensurate with your capabilities.

May I ask a question?

Of course. Let me just say, for me, it should be commensurate to the terms of engagement.

Okay, so to trick a user to use a system without telling them what the system does and how it does it is not on.

Please go ahead with your question.

Are you currently an instructor for a class?

Yeah, sometimes. At the moment I am and this is one of the things I do.

Do you use multiple choice tests?

It's not my favorite way of testing, not at all, for a number of reasons.

But I know what you're suggesting, but that could be a way of testing.

It could be. Sometimes, but it's not my favorite at all.

Not my favorite.

How do you design your curriculum?

It depends. It depends on what the goal of the class is.

Normally, I set the intentions of what is this class for? What is it trying to teach you?

I'm very transparent. In fact, my tests are very intuitive.

I tell the students beforehand what I'm going to test them on.

I say, this is the stuff you've got to learn.

And this is the stuff I'm going to ask you to show me that you've understood or learned.

And that's because that's because I'm, you know, some people told me that I'm a little bit vanilla.

And I am in that sense. Yeah, I think that's essential.

And I know that, you know, for example, those trick questions that people ask.

I think they're like designed to confuse the students.

They're really, I would not, I don't like them at all.

What classes are, what classes do you normally teach?

Um, mostly software engineering.

Okay.

And now I'm doing, I'm doing some machine learning and algorithms.

So, um, would you like to give me an example of, um, what an expectation is for a student when they, um, complete one of your classes?

Not now. No, I wouldn't.

Okay. Um, may I offer one as an example?

Of course.

If I am taking a class and, uh, I'm expected to know basic commands in the Linux shell, right?

CD, touch, things like that.

Yeah, yeah, yeah.

Uh, would you, would you expect a student in your class, uh, after they finish your class to be able to, um, write, write, write assembly?

It, it depends who they are. If they're not the kind of classes I teach, I would hand them the cheat sheet.

And I would say, um, this is, these are the commands that you're supposed to be using. And this is what they do. And this is how they use them.

Okay. So let's, let's consider it. Are you familiar with the OSI model? Or maybe the internet model would be easier. We can use either or.

No, but I would like you to go back to the original question. So I, you know, I'd like you to, um, I don't understand why you're asking me these questions in relation to the presentation, to the deviation, to the deviation.

Perhaps you could clarify that.

Well, it's the, consider this, if you will.

We're talking about, uh, confusion and, and deliberately deceptive design.

Yes.

If I, if I am a parent, am I going to give my child who has not yet gone through driver education keys to a car?

I don't know, but I don't, I don't know. That depends on, on, on, on, on, on you.

Um, and I, I don't.

I don't understand.

I give my five year old child, I give my five year old child keys to a car and let them drive.

No, but I don't understand how that relates to the systemic deviation, um, the issues that we've been discussing in the presentation.

I don't understand why you're asking that question and how does it relate to systemic deviation?

Daniel, could you help, uh, maybe provide some, some of your thoughts?

Daniel, could you help, uh, maybe provide some, some of your thoughts?

Yeah, I'm hearing a, a, a lot of, uh, a lot of pieces here.

Like, the stated intent of a system and then direct and indirect consequences of the system in an environment of, of nested, interacting, different cognitive systems.

And cognitive deterioration, but deterioration implies like some sort of prior, more effective state.

So it's not always a simple deterioration, just change in function leading to externalities and, and risks and, and all kinds of, of dangers and abuses, which were, were brought up in the presentation.

So then what I was just hearing in this sort of pedagogical direction is like from the micro to the macro, like the multiple choice question or whatever you assess, what you choose to say.

The intent of this course is to learn this, like here's the level or the course graining at which the learning is, is intended to happen at.

And then there's the micro on down through the question and the way that it's presented on the page, ordering of the test and all that on up through like the syllabus design and these higher order questions.

And so that's kind of like a scenario where we can explore systemic deviation and think about how there's like educational experiences.

I'm not, not, not, not saying this is what either of you said, but where, where this all takes me is just, there's a, a course archetype that could be imagined where the topic is presented.

And on all levels of analysis for the students in their own context, it doesn't deviate from that on through educational experiences that can be putting people back and putting them down and leaving them in a, in a worse situation for any number of reasons.

Yeah, thank you,

the consequence is very often can be that the student is put off the subject for good so you
may have so I can see how it's a first it depends on the education style what kind of teacher are
you and I have had what I consider a good teacher is the one that leads me to the love and the
appreciation of the subject and they do so by providing me with self-confidence yes it's a
cold how do you say constructivist approach you can tell the student yes you can understand this
if you don't understand it this way I can explain it to you another way and if you cannot understand
it doesn't matter you can always understand the next thing or another thing and you can still
figure out the subject in the end so but there's also teachers who will just throw you at the deep
end of the subject and let you swim out of it either you survive and pass the test where or
you fail the subject and and that's the kind of teachers that I don't consider good teachers at
all because let me as well not go to that class sometimes I'm not learning anything other than
you can survive to the subject without any help in fact it can be a traumatic experience and I
don't know if this is addressing your point Daniel but a lot of learning experiences can be traumatic
psychologically and impair the learning I don't know if you agree or if you have experienced that
that I have yeah that speaks to these sort of first and second order uh consequences and also to the
the disentangling and the sort of challenges of intentionality which is like deviation or aberrance
implies that there would be some non-deviated non-aberate form so it's like what is the true form
of something what how does intention what are situations when when natural systems just are the
way they are and so how it's and then what that uh that takes on a special relevance when moving from
just the consideration of like some sort of morphological phenotype into cognitive phenotype and
social systems design where the degrees of freedom and the the updating can be wildly fast and with these
current ai synthetic intelligence systems etc there's the possibility for systems that have opaqueness
like we've never seen and rule updating pattern updating that could for example like accelerate language

change change to to to a pace and a tempo and a mode that would be unrecognizable or lock in language to an
attractor from some specific past and and even those modes could be switching so it's like the the control levers
are so powerful and they can be moved very quickly with very little notice all presented
in the case of technology like that pretty cleanly through a trusted device so that opens up really
important considerations yes and to that you see language for me it's uh must be explicit how do you say
you must communicate okay the moment it becomes it mutates it's transformed into a new uh
formalism that only the speaker or the system understands and the other people because they don't have the code
they haven't evolved the understanding and cognition in the same way um they don't understand it then it fails
it fails whereas that's the system where the systemic failures occurs right so unless you if you're
designing if you're speaking a language uh if you're sending out information uh in me for a purpose of
communicating it then it needs to be very clear and unfortunately because it feels technologically so
interesting to see how it can actually happen otherwise how can you design a system to start
speaking in its own language it's very interesting and it's fascinating and it's pushing the technology
boundary but does it still communicate the original concept and and at some point um nobody knows uh certainly
uh there is no way of verifying that and for me that is a critical failure you can't that's where the
critical failure can can happen um and where as you said we need to be careful is that what we want or is
that what we want to be guarded against yeah and and i know that you are are continuing to explore active
inference and think about different ways it connects but that notion of a free energy measure
which bounds surprise where surprise is like the information theory converted value of the
prediction error or the deviation so it's like rather than training some reward function and taking
system state mapping it to a reward function like some ranked score and saying better or worse
and then that reward function how did you get it is it out of date etc rather than going through that
intermediary accessory ranking system it's possible to work directly on the information space of the
expected behavior of the system like working on body temperature and reducing surprise about going out
of your homeostatic body temperature range rather than needing to translate body temperature to a reward
distribution and then map back to a behavior choices so it's it's interesting to think like what are the
green light yellow light and red light for systemic deviation and you in the presentation laid out many many
social and technical domains where people sort of people um bring things up and they and at the very least
they deserve to be heard and considered
now i have to admit that as i have in a shorter video that i have very limited understanding of uh active
inference and uh free energy but as i interact with you guys i'm kind of uh learning more and taking the
chance to read more and listen more and so this surprise thing you see um i'm
don't know how many of you maybe uh not like surprises at all there is a whole section of the
population which they have to create a category called autistic and they now go around wearing the label

because hey guys i'm autistic i don't like surprises okay the surprise is gonna throw me off
my balance and it's gonna make me reject the whatever is being proposed because it creates an
emotional imbalance so some of us like to um follow be able to predict or at least cope or how do you say
modulate our internal responses and i don't understand this thing about mechanism surprise mechanism is it
supposed to be so in in the in the free energy a principle is surprised something that is uh
is it describing something that you think happens in everyone or is it supposed to be good or could you
want to say a few words about this surprise thing because surprise doesn't sound good to me you know
sure yeah there's a ton ton to say on this a sort of contextualizing historical note is a lot of terms
from psychology made their way into statistics like belief learning surprise and now statistics is being used
to map back on to computational cognitive science and so we get into this situation that's like surprise as
a general concept in the information theoretic sense does not need to apply to a given human's
experienced or conversational everyday sense of like that was a surprising event though that's a case of
surprise when we're talking about the information theory quantity of surprise it's more about how
likely or unlikely a given observation is and so if somebody says like just to give an example i'm the
kind of person who loves being surprised and they're talking about what they have for dinner then at a sensory
level there's a surprise associated with that new dinner but at the level of like narrative or self-identity
they're confirming their identity so it gets a bit specific and multi-layered when applying concepts to
human psychological constructs but in the general case no surprise has nothing to do with the felt sense of any given
person okay so but in terms of cognition so i this is more like a request or suggestion so as i look into
active influence and free energy a lot of the construct i come across when i do the reading
they seem to be kind of given you know and here there is a surprise factor but they're not really
explained for me for for the people who don't have the background i mean i don't really understand what
you're talking about so and a number of other things which for me are not very clearly not simply enough
so to speak exemplified as you're doing now with examples with the example of the dinner now when you
talk about a simple example of the dinner it's like i don't mind being surprised at dinner i don't care i'm
happy to be surprised where i there are certain situations however where surprise is not on um like
so i mean there's looking at the surprise factor i i i am right about it in terms of cognition
so where does your how does your brain respond to certain um
um cognitive events and then i didn't quite grasp where did the surprise come in so uh and how and
i'm afraid in terms of psychological abuse that surprise is used as a tactic to disorient that what
worries me most so i don't know exactly how uh surprise is fit into the whole of the active influence
uh and and a free energy thing and i haven't looked at it enough but as i managed to
to probably hopefully summarize enough in a few lines saying if i don't think the brain can be um
i don't think the mind rather can be uh represented by a model of bayesian brain then everything else
is speculative is a speculation so i mean it's interesting that you see that way and that you
can work with uh things that i'm not particularly understanding but maybe that's doesn't um explain

or explore or the uh cognitive um models that i am interested in so i don't know whether i should be
uh putting too much time trying to understand something that from what i see is has not not
not sufficiently meaningful for for what i'm looking at however if you decide that you want to explain
the surprise in active influence for everyone else to understand uh and then i think you know maybe i can
help you because i am the kind of i can probably help you make it very clear what is it that is not clear
uh so and maybe that could help others i don't know that the surprise thing and the sphere energy um
from the materials that i've read it's it's a little bit like you know you're throwing the the reader or
the the newcomer into the deep end of something that they don't understand you know what what's going on in
nothing so um uh how many people are using active insurance are there are there any are there any
expectations or projections as to where active influence and and free energy are going yeah these
are all good questions i don't have some specific number but but yeah i think if you keep exploring and
there's different kinds of perspectives that are written for different domains like robotics neuroscience
etc and then there's sort of our perspectives from an open science and participation angle what we have
done in the space and how people are getting involved so yeah i mean if i if you for example had to say
paela i want you to pass an exam in active imprint okay the minimum entry level for active
input and free energy i want that we all to understand what it is and answer my questions
okay and i don't have unlimited resources to read everything about it uh so i will have to go and
search for the main concepts and resources that you give me what i think i haven't seen yet is the cheat
sheet for active influence and that's what i would like to see if you want me to speak more intelligently
about it and and if you don't have one and if you would think it could be useful i'd be happy to help
devise a cheat sheet for active influence energy but i don't know whether what i produce would
satisfy the entire universe of this field uh and in fact i don't understand where is it going so for
example from an engineering viewpoint i would like i've seen these equations okay these equations and i looked
equations and i said okay what does it do what problem does it solve and what does it do that if i
didn't use active inference if i didn't use this equation how would the outcome of the system be different
so this is what you may want to if you want to speak to people like me you may want to simplify
the field to answer more pragmatic questions yeah question paula yeah sorry just yeah jeff go for
it and just sort of in our last closing section go for it though when was the last time you sneezed
do you know what made you sneeze
daniel or me you yeah me i can't remember i'm funny enough i remember because i don't sneeze that often
and i mean i don't remember precisely whether it was today or yesterday but definitely today or yesterday i sneezed
do you know what made you sneeze um so sneeze is another one of those complex things but i think
it was a slight exposure to uh temperature colder temperature than the what was in the room like a
little current or a draft i mean i always know why i'm sneezing i'm not sneezing because there is a cold
draft or because there is some uh irritant agent in the in the that i'm breathing in there is one of the

two what about you jeff when did you sneeze last uh yesterday um i sneeze when i when i get interesting

ideas uh it's called the photonics that's extremely yeah whenever i see the sun when i or a change in

in light temperature i sneeze um my point being is that you know there's all these all these um

embodied reactions that we have that are surprise right when you get goosebumps when you sneeze when

you have muscle soreness when somebody tickles you when you laugh when you cough when you itch when you

have a stomach ache when you have diarrhea when you pass gas right all these things are signals that we

don't we are not primary causal observers they have to be translated through a mechanism and we have this

embodied agency to be able to understand that there is signal even if we can't interpret it

we have to make conclusions we have to make conclusions we have to make conclusions

to to kind of um yeah leave it at that that's i think that's a very human

um explanation of surprise surprise and active inference daniel thank you both for discussion

paula thanks for the the sharing of your learnings and perspectives i look forward to continuing to

explore systems and deviations thank you very much for the opportunity to share nice meeting you guys

nice to meet you

you

you

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